

4.7 Optimization

Motivation: In this section we want to use what we have learned about differential calculus to solve some applied optimization problems. The following outline gives us a way of approaching optimization problems in general.

Guidelines for Optimization Problems

1. Understand the problem
2. Draw a diagram
3. Introduce notation and assign symbols
4. Determine what you are trying to maximum/minimize (Suppose we call this M)
5. Express M in terms of other symbols
6. Use the given information to express M in terms of a single variable.
7. Take the derivative of the function M . Find and verify the existence of an absolute minimum/maximum.

Note: We will not list these steps out when solving problems, but we will keep them in mind.

Important Formulas:

Volumes:

- Box: $V = lwh$
- Sphere: $V = \frac{4}{3}\pi r^3$
- Cylinder: $V = \pi r^2 h$
- Cone: $V = \frac{1}{3}\pi r^2 h$

Surface Areas:

- Box (with all 6 sides): $SA = 2lw + 2lh + 2wh$
- Cylinder: $SA = 2\pi r^2 + 2\pi rh$
- Cone: $SA = \pi r^2 + \pi rs$ (where s is the slant height)

You should also know formulas for areas, circumference, and diameter.

The following theorem will be important for step 7 in the guidelines for optimization problems.

Theorem (First Derivative Test for Absolute Extreme Values):

Suppose that c is a critical number of a continuous function f defined on an interval.

1. If $f'(x) > 0$ for all $x < c$ and $f'(x) < 0$ for all $x > c$, then $f(c)$ is the absolute maximum value of f .
2. If $f'(x) < 0$ for all $x < c$ and $f'(x) > 0$ for all $x > c$, then $f(c)$ is the absolute minimum value of f .

Example 1: A farmer has 2400ft of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?

Example 2: A factor needs to make a cylindrical can which holds $1\text{Liter}=1000\text{cm}^3$ of oil. Find the dimension of the can (i.e., the radius and height) that will minimize the amount of the metal needed to manufacture the can. Round your final answer to the nearest hundredth.

Example 3: Find the point on the parabola $y^2 = 2x$ that is closest to the point $(1, 4)$.

Example 4: An oil refinery is located on the north bank of a straight river that is 2km wide. A pipeline is to be constructed from the refinery to storage tanks located on the south bank of the river 6km east of the refinery. The cost of laying pipe is $\$400/\text{km}$ over land to a point P on the north bank and $\$800/\text{km}$ under the river to the tanks. To minimize the cost of the pipeline, where should P be located? What is the minimum cost of the pipeline?

